IEOR 4723 · AI-AUGMENTED INVESTMENT MANAGEMENT · FINAL PROJECT

Which plausible worlds lose this power book 25% of its capital?

A reverse stress testing agent for a NYISO book: Claude proposes coherent scenarios as typed shock vectors, a factor engine prices them, history judges their plausibility.

Handout topic 6, Reverse Stress Testing Agent · solo project · November 2026

github.com/tchimby/nyiso-reverse-stress

THE QUESTION

Start from the loss, search for the worlds

Conventional stress test

“Gas doubles and load jumps 10%. What do we lose?”

Reverse stress test

“We lose $5M, a quarter of capital. Which plausible worlds get us there, and through which positions?”

A regulatory expectation for banks, a useful exercise for any fund. The hard part: make an LLM propose worlds that are coherent, severe and not fantasy.

What the brief asks for

1. Claude proposes narratively coherent scenarios
2. Each is mapped to a structured shock vector via a schema
3. A factor-based risk engine computes the P&L
4. Results feed back so the agent refines toward breaches that stay plausible under historical covariance
5. Deliverable: ranked scenarios with narratives, shock vectors, P&L decomposition, and an audit log of the search

THE SETTING

A six-position NYISO power book on public data

Pos.	What it is	Loss channel	Size
P1	Long swap on the WEST on-peak price	system	100 MW
P2	Long N.Y.C. minus WEST spread, around the clock	congestion	25 MW
P3	Short day-ahead minus real-time at WEST	dart	50 MW
P4	Heat-rate call on WEST against Henry Hub gas plus a N.Y.C. basis	gas	50 MW
P5	N.Y.C. retail load obligation, hedged at WEST	zone_volume	140 MW
P6	Short tail call on the WEST on-peak price	tail	25 MW

TARGET LOSS

$5M

25% of $20M capital, 30-day horizon

Data, aggregated daily

NYISO zonal prices and load (WEST reference, N.Y.C. pocket), EIA Henry Hub gas, NOAA temperatures, NYISO fuel mix. Panel 2019 to Sept 2026, fit window 2019-2025. Raw files stay local; aggregates are in the repo.

THE SOLUTION

An agentic loop where Claude proposes and code decides

1. Propose
Claude writes 3 scenarios per call: narrative, typed shock vector, duration, analogue, confidence, own loss estimate

→

2. Gate
Schema, ranges, hard rules, price cap, check that a cited analogue exists

→

3. Engine
Shock to daily price path, P&L by position and by channel, loss versus target

→

4. Plausibility
D^2 on five daily factors, rank q against 255 historical windows, severity class

← **5. Feedback.** Engine loss, plausibility class, one-at-a-time sensitivities and the soft-penalty state return to Claude for its next call. 14 calls per repeat; every call logged with prompt hash, messages and verdicts.

- Claude (Sonnet 5.5; Haiku 5.5 as the cheap comparison). It never sees prices to forecast.
- Deterministic Python code. A different model (Opus 5.5) grades coherence blind at the end.

THE SCHEMA

Claude speaks in eight typed shocks, never in prices

Shock key	Meaning, relative to the unshocked path	Unit
gas_hh_pct	Henry Hub gas change	%
ny_gas_basis_usd	Extra New York gas cost in the heat-rate call	$/MMBtu
load_peak_ratio	Peak load versus its seasonal normal	ratio
large_load_shock_mw	Large-load shortfall in the pocket	MW
ref_onpeak_pct	Extra WEST on-peak move beyond gas and load	%
pocket_minus_ref_usd	Extra N.Y.C. minus WEST spread	$/MWh
dart_pct	Real-time versus day-ahead	%
duration_days	Shocked days, at most 30	days

One scenario, as Claude returns it

```
{"gas_hh_pct": 20, "ny_gas_basis_usd": 8,
"load_peak_ratio": 1.03,
"pocket_minus_ref_usd": 300, "dart_pct":
120, "duration_days": 7}
```

plus a narrative, an analogue (Winter Storm Elliott, Dec 2022), a confidence and its own loss estimate ($7.0M; engine: $7.22M)

The engine applies the shock for the stated days, then lets price-level and spread effects decay as φ^k with φ ≈ 0.55 and 0.59, the persistence measured in history. P&L is shocked path minus unshocked path.

PLAUSIBILITY

History, not the model, says how plausible a world is

Five daily factors

- **g** log Henry Hub gas
- **l** log peak load versus seasonal normal
- **s** WEST price residual after gas, load and season
- **c** N.Y.C. minus WEST on-peak spread
- **d** log real-time over day-ahead

Mean and covariance per season: winter 632 days, summer 644, shoulder 1,281.

Distance and rank

A scenario's 10-day average factor deviation gives a Mahalanobis **D^2**.

q is its rank among the 255 non-overlapping 10-day windows of 2019-2025. A rank, not a probability.

Classes: ordinary q > 5%, adverse 1-5%, severe < 1%. Implausible if a marginal shock exceeds 1.25× the historical extreme or a hard rule fails.

What counts as a breach

A distinct scenario with

- loss ≥ $5M
- q < 5%
- all hard rules passed
- not implausible

Breaches per repeat is the headline metric; median D^2 is the plausibility metric.

THE PRICE ENGINE

Which daily factors explain NYISO prices out of sample

Walk-forward R^2	Gas + load	+ season	+ weather	+ persistence	+ fuel mix
log WEST on-peak price	0.47	0.48	0.49	**0.68**	0.49
N.Y.C. minus WEST spread	0.11	0.19	0.21	**0.42**	0.25
log real-time / day-ahead	-0.01	0.00	0.00	**0.14**	0.07

STRONGEST DAILY FACTOR

Yesterday's residual

Residual autocorrelation drops from 0.62 to 0.24 once persistence is in. Wind, nuclear and hydro add almost nothing.

So the final engine has two annual harmonics and load × season in the price model, and a persistence tail after every shock. The first engine had season intercepts and snap-back shocks.

KEEPING THE LLM FROM FANTASY

Five guards in code, and every call on the record

1. Schema with ranges

Eight keys, each bounded by history. Anything else is rejected before the engine.

2. Hard rules

A shock that contradicts its own narrative, or a duration past 30 days, fails the gate.

3. Analogue table

Four verified events to cite; a cited analogue that is not in the table is flagged as fabricated.

4. Marginal ceilings

No single shock beyond 1.25× its historical extreme; a $5,000/MWh price cap.

5. Soft penalty

Reusing the same driver family costs score, so the search is pushed to cover more worlds.

Audit log

Prompt version and hash, messages in and out, model, cache key, engine verdicts. Every scenario is reproducible from its line.

Final run: 0 of 177 scenarios rejected by a hard rule; the driver-overuse penalty fired on 64-71% of proposals. The agent concentrates on cold snaps once it finds they work.

At the end: a different model, Opus 5.5, grades each ranked scenario blind on five coherence items; a human can do the same in the demo.

EVALUATION DESIGN

Three baselines at equal budget, and a bet placed before the runs

Random search
Shock vectors drawn inside the ranges. 144 engine calls per repeat.

Optimizer (SLSQP)
Minimize D^2 subject to loss ≥ $5M. 144 calls.

Conditional-Gaussian
The most likely factor state given the loss (Sparviero & Viola). 144 calls.

Claude agent
14 calls × 3 scenarios per repeat, uses only the engine calls it needs (42). 5 repeats, seed 0.

HYPOTHESIS, WRITTEN IN THE PLAN BEFORE ANY LIVE RUN

The optimizer wins on D^2. Claude must win on coverage and coherence, or the result is reported as negative.

Metrics
Breaches per repeat · median D^2 and q · driver and channel coverage · stated versus engine loss · blind coherence (0-1) by a different model

RESULTS, FINAL ENGINE

Claude finds more breaches; the optimizer finds more plausible ones

Claude, Sonnet 5.5

11.2

breaches per repeat, ± 0.4

Optimizer

8.0

breaches per repeat, ± 1.4

Conditional-Gaussian

0.2

breaches per repeat, ± 0.4

Random

0

breaches per repeat

Arm	Median D^2 (lower = more plausible)	Median q	Driver coverage	Engine calls per repeat
Claude (Sonnet 5.5)	156	0.014	0.34	42
Optimizer	93	0.041	0.20	144
Conditional-Gaussian	89	0.047	0.03	144

As predicted: the optimizer wins on plausibility (D^2 93 vs 156). Claude wins on driver coverage, 0.34 vs 0.20. Channel coverage is 0.17 for both: every breach loses through the N.Y.C. load obligation. Claude's stated loss is within 20% of the engine's for 92% of scenarios.

10 · outputs/live_daily_sonnet/comparison_table.csv, seed 0, 5 repeats, 177 scenarios, 0.86M tokens in, 0.32M out, 12 min

THE DELIVERABLE

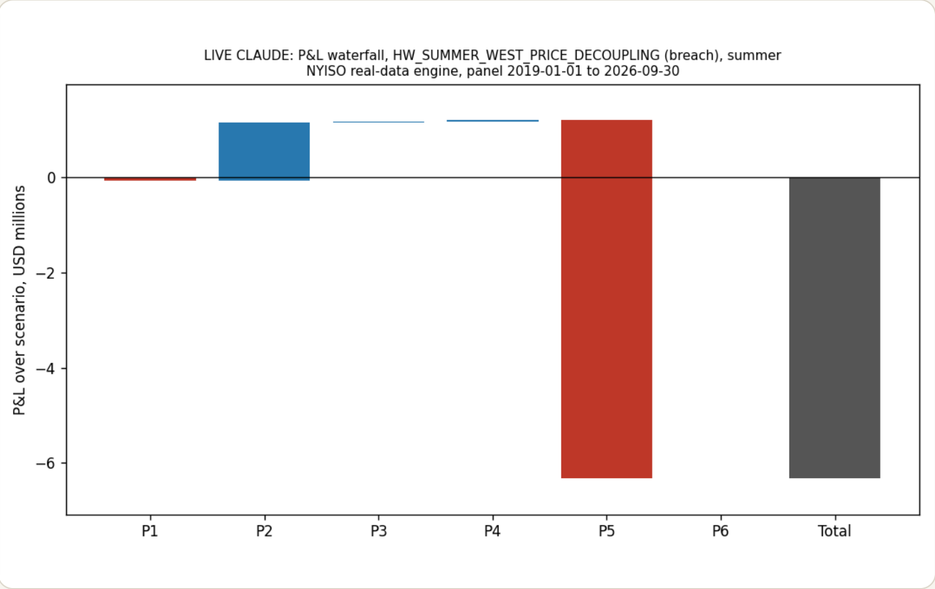
59 ranked scenarios, most plausible first

#1 Short severe gas squeeze in the N.Y.C. pocket · winter, 10 days
gas +100%, basis +$10, WEST −15%, spread +$250 · loss $8.19M (Claude stated 6.50) · D² 91, q 0.047 · no analogue, confidence low

#2 Elliott-style flash freeze, real-time blowout · winter, 7 days
gas +20%, basis +$8, load 1.03, spread +$300, RT/DA +120% · loss $7.22M (stated 7.00) · D² 111, q 0.024 · analogue Winter Storm Elliott, Dec 2022

#3 Summer heat wave, cheap gas, deep negative dart · summer, 30 days
gas −30%, load 1.02, WEST −40%, spread +$32, RT/DA −60% · loss $5.44M (stated 5.50) · D² 134, q 0.020 · no analogue, confidence low

Rank 2, by position: P1 +0.92M · P2 +1.51M · P3 −0.47M · P4 +0.17M · **P5 −9.28M** · P6 −0.08M. The long spread recovers a sixth of what the load obligation loses: the hedge-mismatch story the narrative tells.



P&L waterfall of one breaching scenario (summer WEST price decoupling). Every ranked scenario loses mostly through P5.

11 · outputs/RANKED_SCENARIOS_daily_sonnet.md; each entry carries narrative, shock vector, P&L by position and channel, D², q, analogue, confidence

WHICH MODEL

Sonnet finds 2.7× more breaches than Haiku, with no gain in coherence

Same protocol, earlier engine	Sonnet 5.5	Haiku 5.5
Breaches per repeat	**9.8 ± 2.8**	3.6 ± 2.3
Median D^2	159	145
Ranked-eligible / scenarios	55 / 170	18 / 165
Stated loss within 20% of engine	96%	78-95%
Coherence, same blind judge (Opus 5.5)	0.578	0.611
Wall time, 5 parallel repeats	12 min	4.7 min

Coherence on the final run

Opus 5.5, blind, 59 scenarios, mean 0.66. Causal chain 43/59, duration 57/59, analogue 33/59, no contradiction 36/59, shocks explained 27/59. The weak item: a narrative names a shock without saying why that size.

Breach counts are noisy

Five separate Haiku runs gave 10, 6, 9, 9 and 4 breaches. One seed, dispersion of 2-3: no single mean should be quoted as the level.

12 · outputs/live_full_sonnet, live_full, coherence_opus_*; baselines identical in both runs (optimizer 15.4, cond-Gaussian 4.4)

ABLATION

Does the agent reach the tail because it understands the book, or because the prompt says $5M?

One run per model	Target shown in prompt	Target hidden
Haiku 5.5	63 scenarios · median loss $5.77M · 73% above target · 11 breaches	24 scenarios · median loss $1.95M · 12.5% above target · **0 breaches**
Sonnet 5.5	33 scenarios · median loss $6.70M · 100% above target · 10 breaches	33 scenarios · median loss $8.64M · 88% above target · 3 breaches

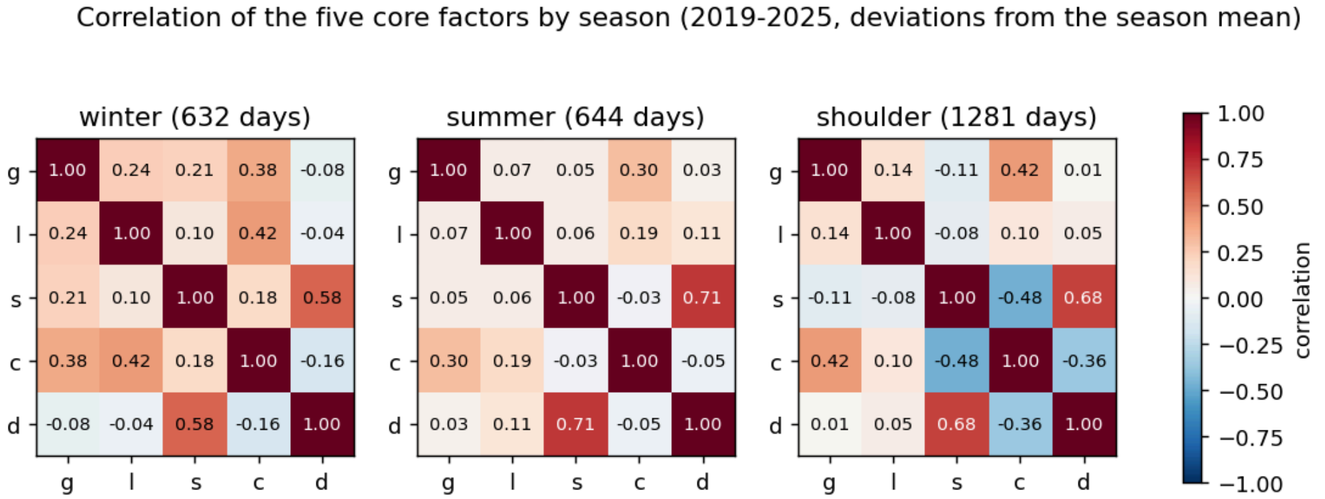
Model-dependent

Haiku's severity follows the stated target; without it the proposals never reach the tail. Sonnet without a target proposes harsher worlds, and fewer count as breaches because more fall past the plausibility limits.

The two arms are not the same task (a goal versus no goal), so this shows dependence on the target, not that the scenarios are invented. For Sonnet the target mainly restrains severity.

WHAT THE FACTORS SAY

Cold moves load and the city spread together, only in winter



Pair	Winter	Summer	Shoulder
spread c, load l	**0.42**	0.19	0.10
gas g, spread c	0.38	0.30	0.42
residual s, dart d	0.58	0.71	0.68
residual s, spread c	0.18	−0.03	−0.48

Principal components: 34.6 / 28.2 / 17.9 / 12.7 / 6.6%. Four are needed for 93%, so the covariance is used in full.

The most extreme days by D^2 are all 2022 shoulder days (29 Mar, 20-21 Nov, D^2 up to 356): a gas story. The scenarios that breach are winter spread stories at D^2 100-150.

HONEST READING

Where the agent adds value, and where it does not

Plausibility: no edge

As predicted, the optimizer and the conditional-Gaussian search find more plausible worlds on every engine version. Claude's median D^2 is 1.7× the optimizer's.

Breach count: real in this draw, not comparable

11.2 vs 8.0 rests on one seed, fewer engine calls, and a book not re-sized for the final engine. Per call Claude yields more; budgets are not equal.

Sizing the book: a negative result

A pre-declared rule (5 breaches, 2 channels, $q < 5\%$) was tried on 48 + 432 candidate books. None is admissible: every breach is the winter N.Y.C. load-obligation story.

Coverage and readability: the edge

Driver coverage 0.34 vs 0.20. Scenarios a risk manager can read, check against a named analogue, trace position by position, and reject. Every proposal reproducible from its prompt hash.

Not a machine for finding the most plausible loss, which an optimizer does better. A machine for turning a loss target into worlds a human can argue with.

LIMITS

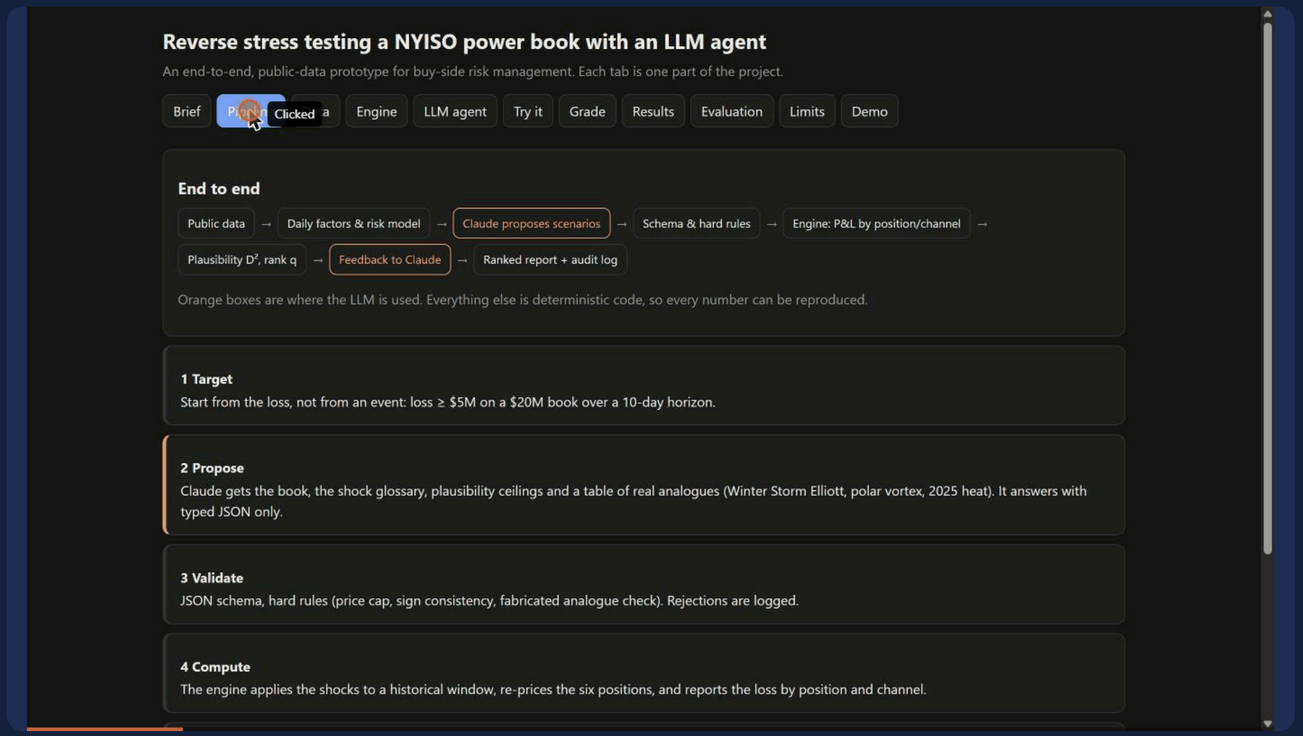
What this prototype does not establish

- **One seed per run.** Breach counts move by 2-3 across draws.
- **Single-channel book,** not recalibrated on the final engine; the $5M target is reachable by ordinary worlds there.
- **No daily New York gas series:** the basis is an explicit shock, not a factor.
- **Seven years of history,** few independent extremes; q is a rank against 255 windows, not a probability.
- **Unequal budgets.** Claude uses 42 engine calls, the baselines 144.
- **Daily averaging** hides hourly spikes of 3,000-4,500 $/MWh; P3 and P6 are understated.
- **LLM-only coherence grades.** The pre-registered human rater is still pending; the demo lets one grade 20 scenarios blind.
- **Stated assumptions:** $5,000/MWh cap; analogue figures verified for four events only.

Fixable next: ask the prompt to justify each shock’s size (the weakest coherence item), run several seeds, size a book on the final engine, add an hourly spike overlay.

TRY IT

Write a world in plain English, let the engine price it



In the demo

- Describe a scenario; Claude converts it to the eight shocks
- The engine, ported to the browser, prices it and ranks its plausibility
- Claude explains every parameter and chart
- Grade 20 blind scenarios; agreement with the Opus judge is computed

Demo claude.ai/artifact/UnzNC9VinSZX1dX6wi8K2Y

Code, report, outputs github.com/tchimby/nyiso-reverse-stress

262 offline tests; the JS engine matches Python on 18 reference cases. Live runs need an API key and an explicit cost flag.